A1

For any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers with sum $s_A = a_1+a_2+a_3+a_4$, let p_A denote the number of pairs (i, j) with $1 \le i < j \le 4$ for which $a_i + a_j$ divides s_A. Among all sets of four distinct positive integers, determine those sets A for which p_A is maximal.

Answer. The sets A for which p_A is maximal are the sets the form $\{d, 5d, 7d, 11d\}$ and $\{d, 11d, 19d, 29d\}$, where d is any positive integer. For all these sets p_A is 4.

Solution. Firstly, we will prove that the maximum value of p_A is at most 4. Without loss of generality, we may assume that $a_1 < a_2 < a_3 < a_4$. We observe that for each pair of indices (i, j) with $1 \le i < j \le 4$, the sum $a_i + a_j$ divides s_A if and only if $a_i + a_j$ divides $s_A - (a_i + a_j) = a_k + a_l$, where k and l are the other two indices. Since there are 6 distinct pairs, we have to prove that at least two of them do not satisfy the previous condition. We claim that two such pairs are (a_2, a_4) and (a_3, a_4). Indeed, note that $a_2 + a_4 > a_1 + a_3$ and $a_3 + a_4 > a_1 + a_2$. Hence $a_2 + a_4$ and $a_3 + a_4$ do not divide s_A. This proves $p_A \le 4$.

Now suppose $p_A = 4$. By the previous argument we have

$$\begin{array}{lcl} a_1 + a_4 \mid a_2 + a_3 & \text{and} & a_2 + a_3 \mid a_1 + a_4, \\ a_1 + a_2 \mid a_3 + a_4 & \text{and} & a_3 + a_4 \nmid a_1 + a_2, \\ a_1 + a_3 \mid a_2 + a_4 & \text{and} & a_2 + a_4 \nmid a_1 + a_3. \end{array}$$

Hence, there exist positive integers m and n with $m > n \ge 2$ such that

$$\begin{cases} a_1 + a_4 = a_2 + a_3 \\ m(a_1 + a_2) = a_3 + a_4 \\ n(a_1 + a_3) = a_2 + a_4. \end{cases}$$

Adding up the first equation and the third one, we get $n(a_1 + a_3) = 2a_2 + a_3 - a_1$. If $n \ge 3$, then $n(a_1 + a_3) > 3a_3 > 2a_2 + a_3 > 2a_2 + a_3 - a_1$. This is a contradiction. Therefore $n = 2$. If we multiply by 2 the sum of the first equation and the third one, we obtain

$$6a_1 + 2a_3 = 4a_2,$$

while the sum of the first one and the second one is

$$(m + 1)a_1 + (m - 1)a_2 = 2a_3.$$

Adding up the last two equations we get

$$(m + 7)a_1 = (5 - m)a_2.$$

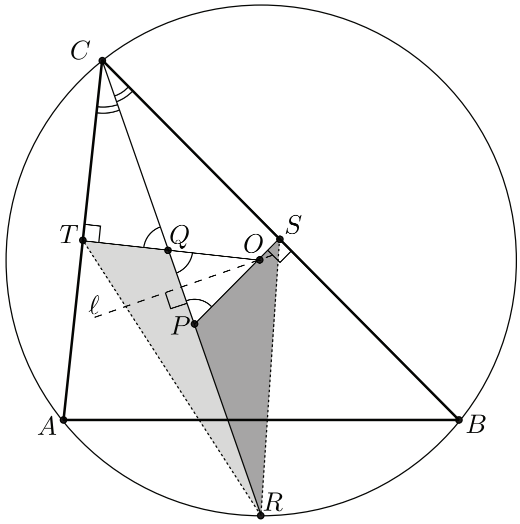
It follows that $5 - m \geq 1$, because the left-hand side of the last equation and a_2 are positive. Since we have $m > n = 2$, the integer m can be equal only to either 3 or 4. Substituting $(3, 2)$ and $(4, 2)$ for (m, n) and solving the previous system of equations, we find the families of solutions $\{d, 5d, 7d, 11d\}$ and $\{d, 11d, 19d, 29d\}$, where d is any positive integer.

Geometry

G1. In triangle ABC, the angle bisector at vertex C intersects the circumcircle and the perpendicular bisectors of sides BC and CA at points R, P, and Q, respectively. The midpoints of BC and CA are S and T, respectively. Prove that triangles RQT and RPS have the same area.

(Czech Republic)

Solution 1. If $AC = BC$ then triangle ABC is isosceles, triangles RQT and RPS are symmetric about the bisector CR and the statement is trivial. If $AC \neq BC$ then it can be assumed without loss of generality that $AC < BC$.



Denote the circumcenter by O. The right triangles CTQ and CSP have equal angles at vertex C, so they are similar, $\angle CPS = \angle CQT = \angle OQP$ and

$$\frac{QT}{PS} = \frac{CQ}{CP}. \tag{1}$$

Let ℓ be the perpendicular bisector of chord CR; of course, ℓ passes through the circumcenter O. Due to the equal angles at P and Q, triangle OPQ is isosceles with $OP = OQ$. Then line ℓ is the axis of symmetry in this triangle as well. Therefore, points P and Q lie symmetrically on line segment CR,

$$RP = CQ \quad \text{and} \quad RQ = CP. \tag{2}$$

Triangles RQT and RPS have equal angles at vertices Q and P, respectively. Then

$$\frac{\operatorname{area}(RQT)}{\operatorname{area}(RPS)} = \frac{\frac{1}{2} \cdot RQ \cdot QT \cdot \sin \angle RQT}{\frac{1}{2} \cdot RP \cdot PS \cdot \sin \angle RPS} = \frac{RQ}{RP} \cdot \frac{QT}{PS}.$$

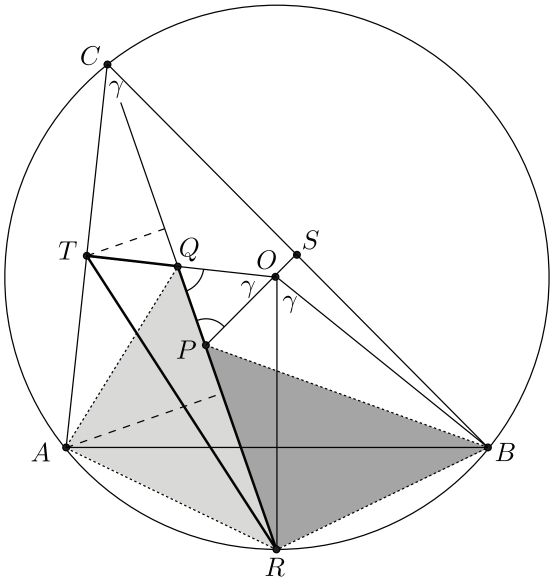
Substituting (1) and (2),

$$\frac{\operatorname{area}(RQT)}{\operatorname{area}(RPS)} = \frac{RQ}{RP} \cdot \frac{QT}{PS} = \frac{CP}{CQ} \cdot \frac{CQ}{CP} = 1.$$

Hence, $\operatorname{area}(RQT) = \operatorname{area}(RSP)$.

Solution 2. Assume again $AC < BC$. Denote the circumcenter by O, and let γ be the angle at C. Similarly to the first solution, from right triangles CTQ and CSP we obtain that $\angle OPQ = \angle OQP = 90^\circ - \frac{\gamma}{2}$. Then triangle OPQ is isosceles, $OP = OQ$ and moreover $\angle POQ = \gamma$.

As is well-known, point R is the midpoint of arc AB and $\angle ROA = \angle BOR = \gamma$.



Consider the rotation around point O by angle γ. This transform moves A to R, R to B and Q to P; hence triangles RQA and BPR are congruent and they have the same area.

Triangles RQT and RQA have RQ as a common side, so the ratio between their areas is

$$\frac{\text{area}(RQT)}{\text{area}(RQA)} = \frac{d(T, CR)}{d(A, CR)} = \frac{CT}{CA} = \frac{1}{2}.$$

($d(X, YZ)$ denotes the distance between point X and line YZ).

It can be obtained similarly that

$$\frac{\text{area}(RPS)}{\text{area}(BPR)} = \frac{CS}{CB} = \frac{1}{2}.$$

Now the proof can be completed as

$$\text{area}(RQT) = \frac{1}{2}\,\text{area}(RQA) = \frac{1}{2}\,\text{area}(BPR) = \text{area}(RPS).$$

C2

Suppose that 1000 students are standing in a circle. Prove that there exists an integer k with $100 \le k \le 300$ such that in this circle there exists a contiguous group of $2k$ students, for which the first half contains the same number of girls as the second half.

Solution. Number the students consecutively from 1 to 1000. Let $a_i = 1$ if the ith student is a girl, and $a_i = 0$ otherwise. We expand this notion for all integers i by setting $a_{i+1000} = a_{i-1000} = a_i$. Next, let

$$S_k(i) = a_i + a_{i+1} + \cdots + a_{i+k-1}.$$

Now the statement of the problem can be reformulated as follows:

There exist an integer k with $100 \le k \le 300$ and an index i such that $S_k(i) = S_k(i+k)$.

Assume now that this statement is false. Choose an index i such that $S_{100}(i)$ attains the maximal possible value. In particular, we have $S_{100}(i-100) - S_{100}(i) < 0$ and $S_{100}(i) - S_{100}(i+100) > 0$, for if we had an equality, then the statement would hold. This means that the function $S(j) - S(j+100)$ changes sign somewhere on the segment $[i-100, i]$, so there exists some index $j \in [i-100, i-1]$ such that

$$S_{100}(j) \le S_{100}(j+100) - 1, \quad \text{but} \quad S_{100}(j+1) \ge S_{100}(j+101) + 1. \tag{1}$$

Subtracting the first inequality from the second one, we get $a_{j+100} - a_j \ge a_{j+200} - a_{j+100} + 2$, so

$$a_j = 0, \quad a_{j+100} = 1, \quad a_{j+200} = 0.$$

Substituting this into the inequalities of (1), we also obtain $S_{99}(j+1) \le S_{99}(j+101) \le S_{99}(j+1)$, which implies

$$S_{99}(j+1) = S_{99}(j+101). \tag{2}$$

Now let k and ℓ be the least positive integers such that $a_{j-k} = 1$ and $a_{j+200+\ell} = 1$. By symmetry, we may assume that $k \ge \ell$. If $k \ge 200$ then we have $a_j = a_{j-1} = \cdots = a_{j-199} = 0$, so $S_{100}(j-199) = S_{100}(j-99) = 0$, which contradicts the initial assumption. Hence $\ell \le k \le 199$.

Finally, we have

$$S_{100+\ell}(j-\ell+1) = (a_{j-\ell+1} + \cdots + a_j) + S_{99}(j+1) + a_{j+100} = S_{99}(j+1) + 1,$$
$$S_{100+\ell}(j+101) = S_{99}(j+101) + (a_{j+200} + \cdots + a_{j+200+\ell-1}) + a_{j+200+\ell} = S_{99}(j+101) + 1.$$

Comparing with (2) we get $S_{100+\ell}(j-\ell+1) = S_{100+\ell}(j+101)$ and $100 + \ell \le 299$, which again contradicts our assumption.

Comment. It may be seen from the solution that the number 300 from the problem statement can be replaced by 299. Here we consider some improvements of this result. Namely, we investigate which interval can be put instead of $[100, 300]$ in order to keep the problem statement valid.

First of all, the two examples

$$\underbrace{1,1,\dots,1}_{167},\underbrace{0,0,\dots,0}_{167},\underbrace{1,1,\dots,1}_{167},\underbrace{0,0,\dots,0}_{167},\underbrace{1,1,\dots,1}_{167},\underbrace{0,0,\dots,0}_{165}$$

and

$$\underbrace{1,1,\dots,1}_{249},\underbrace{0,0,\dots,0}_{251},\underbrace{1,1,\dots,1}_{249},\underbrace{0,0,\dots,0}_{251}$$

show that the interval can be changed neither to $[84, 248]$ nor to $[126, 374]$.

On the other hand, we claim that this interval can be changed to $[125, 250]$. Note that this statement is invariant under replacing all 1's by 0's and vice versa. Assume, to the contrary, that there is no admissible $k \in [125, 250]$. The arguments from the solution easily yield the following lemma.

Lemma. Under our assumption, suppose that for some indices $i < j$ we have $S_{125}(i) \le S_{125}(i+125)$ but $S_{125}(j) \ge S_{125}(j+125)$. Then there exists some $t \in [i, j-1]$ such that $a_t = a_{t-1} = \cdots = a_{t-125} = 0$ and $a_{t+250} = a_{t+251} = \cdots = a_{t+375} = 0$. $\square$

Let us call a segment $[i, j]$ of indices a *crowd*, if (a) $a_i = a_{i+1} = \cdots = a_j$, but $a_{i-1} \ne a_i \ne a_{j+1}$, and (b) $j - i \ge 125$. Now, using the lemma, one can get in the same way as in the solution that there exists some crowd. Take all the crowds in the circle, and enumerate them in cyclic order as $A_1, \dots, A_d$. We also assume always that $A_{s+d} = A_{s-d} = A_s$.

Consider one of the crowds, say A_1. We have $A_1 = [i, i+t]$ with $125 \le t \le 248$ (if $t \ge 249$, then $a_i = a_{i+1} = \cdots = a_{i+249}$ and therefore $S_{125}(i) = S_{125}(i+125)$, which contradicts our assumption). We may assume that $a_i = 1$. Then we have $S_{125}(i+t-249) \le 125 = S_{125}(i+t-124)$ and $S_{125}(i) = 125 \ge S_{125}(i+125)$, so by the lemma there exists some index $j \in [i+t-249, i-1]$ such that the segments $[j-125, j]$ and $[j+250, j+375]$ are contained in some crowds.

Let us fix such j and denote the segment $[j+1, j+249]$ by B_1. Clearly, $A_1 \subseteq B_1$. Moreover, B_1 cannot contain any crowd other than A_1 since $|B_1| = 249 < 2 \cdot 126$. Hence it is clear that $j \in A_d$ and $j + 250 \in A_2$. In particular, this means that the genders of A_d and A_2 are different from that of A_1.

Performing this procedure for every crowd A_s, we find segments $B_s = [j_s+1, j_s+249]$ such that $|B_s| = 249$, $A_s \subseteq B_s$, and $j_s \in A_{s-1}$, $j_s + 250 \in A_{s+1}$. So, B_s covers the whole segment between A_{s-1} and A_{s+1}, hence the sets $B_1, \dots, B_d$ cover some 1000 consecutive indices. This implies $249d \ge 1000$, and $d \ge 5$. Moreover, the gender of A_i is alternating, so d is even; therefore $d \ge 6$.

Consider now three segments $A_1 = [i_1, i'_1]$, $B_2 = [j_2+1, j_2+249]$, $A_3 = [i_3, i'_3]$. By construction, we have $[j_2-125, j_2] \subseteq A_1$ and $[j_2+250, j_2+375] \subseteq A_3$, whence $i_1 \le j_2 - 125$, $i'_3 \ge j_2 + 375$. Therefore $i'_3 - i_1 \ge 500$. Analogously, if $A_4 = [i_4, i'_4]$, $A_6 = [i_6, i'_6]$ then $i'_6 - i_4 \ge 500$. But from $d \ge 6$ we get $i_1 < i'_3 < i_4 < i'_6 < i_1 + 1000$, so $1000 > (i'_3 - i_1) + (i'_6 - i_4) \ge 500 + 500$. This final contradiction shows that our claim holds.

One may even show that the interval in the statement of the problem may be replaced by $[125, 249]$ (both these numbers cannot be improved due to the examples above). But a proof of this fact is a bit messy, and we do not present it here.